A Rolling-Origin Benchmark for Forecasting German Household Overnight Deposit Volumes

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Abstract

Liquidity management requires accurate forecasts of non-maturity deposits (NMDs), yet optimal modeling approaches remain underdeveloped. This study establishes a controlled rolling-origin benchmark comparing naive baselines, regularized linear regression (Ridge), gradient-boosted trees (XGBoost), a feedforward neural network, and a Temporal Fusion Transformer. Using German household overnight deposit volumes over a 100-day horizon (2009–2025), all models evaluate across four distinct interest-rate regimes under identical information sets, 100-day lookback windows, and tuning budgets. Ridge regression achieved the highest accuracy (MAE = 9.36 Bn. €; relMAE = 0.44) and was the only model with a positive out-of-sample R^2 (0.178). Non-linear architectures failed to beat a simple one-step drift benchmark (relMAE = 0.60). Increased model complexity does not translate into higher forecasting accuracy, highlighting regularized linear models as a baseline for bank asset-liability management.

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1 Introduction

The Basel Committee on Banking Supervision (Basel Committee on Banking Supervision, 2016) defines non-maturity deposits (NMDs) as liabilities without contractual maturity where depositors retain immediate withdrawal rights. Because interest paid on NMDs is below market rates and banks adjust deposit rates flexibly, NMDs provide stable, low-cost funding. In the German banking sector, household overnight deposits account for over 60% of retail funding (Schlüter et al., 2015), making their predictable management critical for bank liquidity management, balance-sheet planning, and regulatory compliance (European Central Bank, 2011). However, because depositor behavior is driven by stochastic cash-flow needs, forecasting volume trajectories remains methodologically challenging, and literature lacks consensus on the optimal modeling approach (Winkler, 2025; Marena et al., 2022).

This study addresses this gap by comparing the forecasting NMD volumes. Because bank-level NMD data are protected, our empirical analysis uses publicly observable German household overnight deposit data as an aggregate as seen in Section 6.6.

The study makes three main contributions. First, it examines whether model complexity leads to higher accuracy under identical information sets and hyperparameter tuning budgets across all models. Second, it investigates how interest-rate, macroeconomic, and market covariates affect forecasting accuracy. Third, it evaluates model stability across four distinct interest-rate regimes using absolute and scale-free error measures.

The remainder of this paper is structured as follows. To position our benchmark within NMD literature and forecasting complexity debates, Section 2 reviews related research. To handle non-stationarity, daily frequency alignment, and delayed pass-through, Section 3 details data preprocessing, PCHIP interpolation, and lag-augmentation. Section 4 formalizes the multi-horizon framework, model specifications, and rolling-origin evaluation protocol. Section 5 reports empirical accuracy across four macroeconomic regimes. After that Section 6 analyzes architectural failure modes, limitations, and managerial implications. Finally, Section 7 concludes and outlines future research approaches.

Research Questions:

RQ1 How accurately can statistical and machine-learning models forecast German household overnight deposit volumes over a 100-day horizon, measured in absolute terms and relative to a naive benchmark?

RQ2 Do nonlinear machine-learning models outperform linear and naive baselines under the same rolling-origin framework and tuning budget?

RQ3 Which covariate specifications (market indicators, interest rates, or macroeconomics) maximize forecast accuracy and stability across regimes?

2 Related Work

Modelling non-maturity deposits Most NMD literature uses deposit volumes as a modelling input rather than the forecasting target. Jarrow and van Deventer (1998) model deposit balances and rates jointly under an arbitrage-free framework, while Hutchison and Pennacchi (1996) examine volume sensitivity under imperfect competition. O'Brien (2000) demonstrates slow balance convergence using partial-adjustment models, motivating strong target autocorrelation. Another way is to translates volume dynamics into replicating-portfolio techniques for asset–liability management (Frauendorfer and Schürle, 2003; Kalkbrener and Willing, 2004; Blöchlinger, 2015; Nyström, 2008), an approach embedded in supervisory maturity caps (Basel Committee on Banking Supervision, 2016; European Banking Authority, 2022).

Machine learning for NMD volumes Winkler (2025) compares a Lévy-driven Ornstein–Uhlenbeck process against an LSTM network, providing interpolation procedures we adopt in Section 3.2. Marena et al. (2022) also apply ML to NMDs. However, both studies evaluate on a single out-of-sample window, which fails to separate model superiority from window-specific fit (Hewamalage et al., 2021).

Complexity in Forecasting Other literature on forecasting already showed that simple statistical and hybrid models systematically beat pure ML methods against complexity (Makridakis et al., 2020, 2022). Zeng et al. (2023) demonstrate that single linear layers match or exceed complex Transformers on long-horizon benchmarks, arguing self-attention disrupts temporal ordering. Hewamalage et al. (2021) show neural network gains rarely survive rigorous rolling-origin protocols with matched tuning effort.

Positioning. Table 1 positions this paper. We are not aware of any study benchmarking naive, regularized linear, tree-based, and attention models on NMD volumes under common rolling origins, identical information sets, matched tuning budgets, and scale-free metrics.

Table 1: Positioning of this paper relative to related work.

Reference	Modeling approach	Multiple origins	Matched tuning
Jarrow and van Deventer (1998)	Arbitrage-free valuation	✗	—
O'Brien (2000)	Partial-adjustment model	✗	—
Frauendorfer and Schürle (2003)	Replicating portfolios	✗	—
Marena et al. (2022)	Machine learning for NMDs	✗	✗
Winkler (2025)	Lévy OU process vs. LSTM	✗	✗
Makridakis et al. (2020)	Large-scale M4 competition	✓	✓
Zeng et al. (2023)	Linear layer vs. Transformers	✓	✓
This paper	Naive, Ridge, XGBoost, FFN, TFT	✓	✓

3 Data and Preprocessing

3.1 Target Variable and Explanatory Variables

Because bank NMD data are private, we use publicly available end-of-period household overnight deposit volumes from the Deutsche Bundesbank (Deutsche Bundesbank, 2025a) (January 2003 – September 2025, measured in EUR billions). Household overnight deposits constitute a major subset of NMDs and account for a great amount of bank funding (Deutsche Bundesbank, 2024).

We select six macro-financial covariates based on economic logic:

- **€STR Rate:** Sourced from the ECB (European Central Bank, 2025b) at daily frequency; reflects interbank money market borrowing costs that drive retail rate adjustments.
- **Geopolitical Risk (GPRC_DEU):** Monthly index (Caldara and Iacoviello, 2022) capturing geopolitical uncertainty, driving precautionary liquidity hoarding.
- **MoM Inflation:** Monthly CPI percentage change from Destatis (Destatis, 2025); influences household real disposable income and savings capacity.
- **DAX Index:** Daily closing prices via yfinance (Aroussi, 2025; Yahoo Finance, 2026); reflects German equity market confidence and asset allocation alternatives.
- **10Y Bond Yield:** Business-day German sovereign yield (Deutsche Bundesbank, 2025b); measures opportunity costs of holding liquid cash versus default-free fixed income.

- **Retail Deposit Rate:** Monthly MFI interest rate (Deutsche Bundesbank, 2025a); measures direct returns on retail overnight holdings.

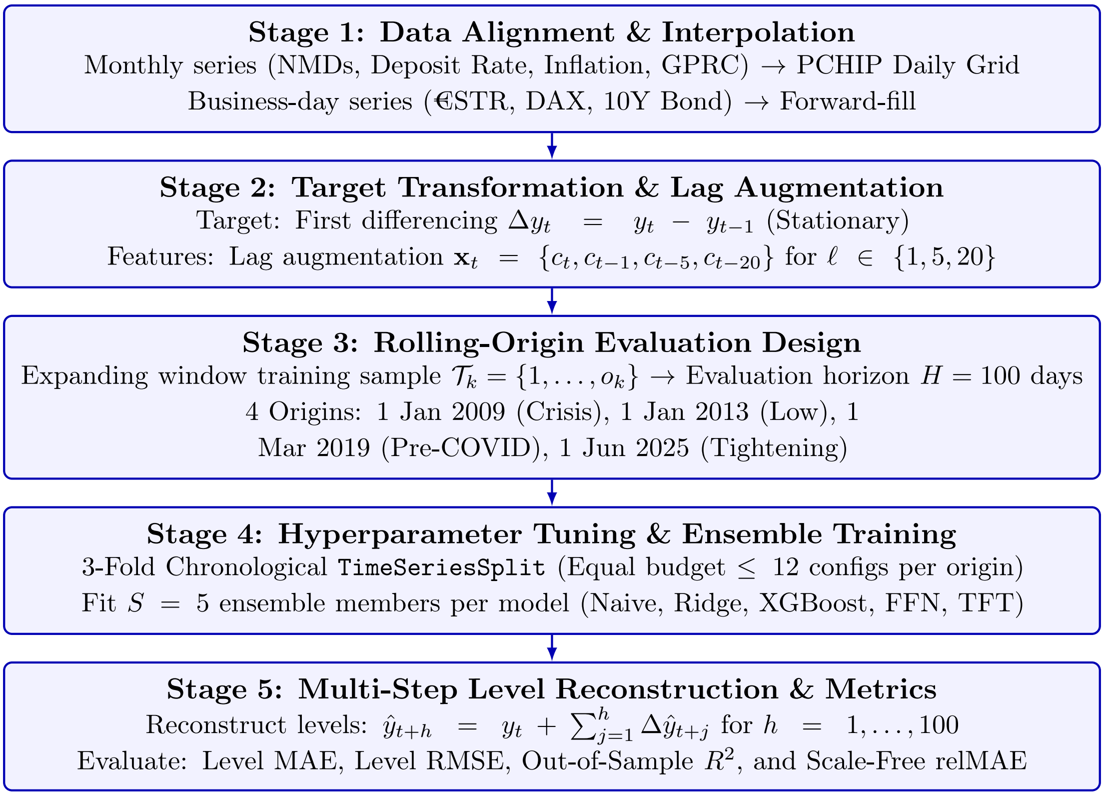

Figure 1: End-to-end forecasting pipeline and rolling-origin evaluation setup.

3.2 Data Alignment and Transformation

Monthly variables (volume, deposit rate, inflation, geopolitical risk) are interpolated to a daily grid using Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) interpolation (Winkler, 2025, Ch. 5.2). PCHIP preserves local monotonicity and avoids artificial overshoots between monthly points (Fritsch and Carlson, 1980).

To reconstruct €STR prior to its October 2019 launch, Pre-€STR (European Central Bank, 2025a) is used for March 2017 – September 2019, and synthetic reconstruction for earlier observations (Winkler, 2025, Ch. 5.2). For business-day series (€STR, DAX, 10Y Bond), holiday gaps are forward-filled.

Figure 2 shows all standardized variables over the sample period.

3.3 Lagged Covariates

Economic covariates do not act on deposit behaviour instantaneously, this introduces a delay in the pass through between the policy rate and retail deposit rates, therefore households reallocate balances only gradually once that pass-through has occurred; the partial-adjustment evidence of O'Brien (2000) points in the same direction. A specification that contains only the contemporaneous value $\mathbf{x}_t$ implicitly assumes an immediate effect and therefore cannot represent this mechanism, even if it is present in the data.

We therefore augment every covariate set with lagged copies of each covariate at $\ell \in \{1, 5, 20\}$ trading days, corresponding to a one-day, one-week, and roughly one-month delay. For a covariate set $\mathcal{C}$, the feature vector at time t is

$$\mathbf{x}_t = (\{c_t\}_{c\in\mathcal{C}},\ \{c_{t-1}\}_{c\in\mathcal{C}},\ \{c_{t-5}\}_{c\in\mathcal{C}},\ \{c_{t-20}\}_{c\in\mathcal{C}}) \tag{1}$$

overview.png

Figure 2: Time series of all variables (Jan 2003 – Sep 2025) after daily PCHIP interpolation and standardization.

The contemporaneous value is retained alongside the lags so that immediate and delayed effects can both be represented and the model, rather than the analyst, selects between them. Shifting is performed positionally on the trading-day index, so a lag of five denotes five trading days rather than five calendar days; the first $\max_\ell \ell = 20$ rows of the sample are lost.

Because the lookback window itself spans 100 trading days (Section 4.1), the explicit lag features are strictly nested within the information the models already receive. Their purpose is not to extend the reach of the information set but to make delayed relationships directly representable: a flattened window presents the same value at 100 distinct positions, and a lag feature states explicitly which displacement is economically meaningful, which is a substantially easier learning problem for models without an internal notion of temporal displacement. Lag augmentation therefore applies to Ridge, XGBoost and the FFN; the TFT processes the covariate history sequentially through its encoder and does not require explicit lag columns.

3.4 Target Transformation

As evident from Figure 2, the deposit-volume level exhibits a clear upward trend throughout the timeline, indicating nonstationarity in the mean. First differencing is therefore applied to obtain a stationary target series and to limit accumulating directional errors (Box et al., 2015), as illustrated in Figure 3. Second differencing was rejected due to over-differencing.

figures/diff_volume.png

Figure 3: First difference of the deposit volume, Δy_t, in EUR billions per day (vertical axis) over the full sample period January 2003 – September 2025 (horizontal axis). The series fluctuates around zero throughout, which is consistent with stationarity after differencing. Increased volatility from 2022 onward coincides with the ECB interest-rate tightening cycle.

After interpolation to daily frequency, the dataset comprises 8,276 daily observations covering January 2003 to September 2025. Because the target is published monthly, these daily rows correspond to approximately 273 independent monthly releases. The effective sample size is therefore roughly thirty times smaller than the nominal number of rows, a point we return to in Section 6.6.

4 Forecasting Models

4.1 Forecasting Framework

We define y_t as the volume of German household overnight deposits at time t, and $\mathbf{x}_t$ as the vector of our predictors and their lagged values from Equation (2). The objective is to forecast deposit volumes over a 100-day horizon ($H = 100$) using the information available during the preceding 100 days ($L = 100$). The reasoning for $H = 100$ is because it accounts for roughly one quarter of trading days, therefore relevant for the operational liquidity management's planning cycle.

At time of forecast t the information set contains the historical deposit volume and the selected covariates over the period $t-L+1,\ldots,t$. Each model produces the complete vector of predicted changes for the following $H = 100$ days:

$$\widehat{\boldsymbol{\Delta y}}_{t+1:t+H} = f(y_{t-L+1:t}, \mathbf{x}_{t-L+1:t}), \qquad L = 100, \quad H = 100. \tag{2}$$

A direct multi-output forecasting strategy is used for this study. Each model estimates all 100 future changes simultaneously rather than recursively one day at a time. The predicted changes are then converted back into deposit-volume levels according to

$$\widehat{y}_{t+h} = y_t + \sum_{j=1}^{h} \widehat{\Delta y}_{t+j}, \qquad h = 1,\ldots,H. \tag{3}$$

Only information available up to time t is used for model estimation and for fitting the feature scaler, which is a robust scaler estimated on the training window alone. Actual covariate values within the 100 forecast days are never used. For the TFT, covariates are held constant at their most recently observed value over the horizon, while date-based variables that are known in advance are supplied for the entire forecast horizon.

4.2 Model Specifications

Naive benchmarks. The one-step drift benchmark extrapolates the latest daily change,

$$\hat{y}_{t_0+h} = y_{t_0} + h\Delta y_{t_0},$$

whereas the constant-level benchmark sets $\hat{y}_{t_0+h} = y_{t_0}$.

Linear regression (Ridge). A multi-output ridge regression maps the flattened 100-day lookback window to the 100-step forecast vector, regularized by an ℓ_2 penalty α. As a linear model, Ridge yields directly inspectable coefficients. Also, the flattened feature vector is high-dimensional and strongly collinear, so it makes shrinkage adequate.

XGBoost. Gradient-boosted regression trees are trained on the same flattened input representation (Chen and Guestrin, 2016), with maximum depth, learning rate and subsample ratio selected by chronological time-series cross-validation. Boosted trees can capture non-linear covariate interactions that Ridge cannot (Hastie et al., 2009, Ch. 10), while remaining moderately interpretable.

Feedforward neural network (FFN). The FFN is a multilayer perceptron with ReLU activations that predicts the 100-step output vector directly. Its layer configuration and ℓ_2 penalty are selected using the same chronological cross-validation procedure and tuning budget as Ridge and XGBoost (Section 4.3). Training uses the Adam optimizer and early stopping on the final 15% of the training window, with a patience of 20 epochs and a maximum of 500 epochs. The chronological validation split preserves the time ordering of the observations, although adjacent 100-day sequences may still overlap.

Temporal Fusion Transformer (TFT). The TFT explicitly distinguishes between known future inputs (calendar features: weekday, month-end indicator, day-of-month) and unknown future inputs (the six macro-financial covariates). The encoder length is set to $L = 100$ to match the other models, and the model is trained with early stopping on validation loss with a patience of five validation checks. Despite being the most complex model considered, its attention mechanism was explicitly designed to offer a degree of interpretability that purely black-box models lack.

Each model is fitted $S = 5$ times to form an ensemble. Ridge, FFN and XGBoost use bootstrap-resampled training sequences together with different random seeds; repeated TFT fits differ in the random seed only. Ensemble members are combined by averaging their forecasts.

4.3 Tuning Protocol

A comparison of "complex" against "simple" models is only meaningful if the models receive comparable optimisation effort. We therefore impose a single tuning budget of at most twelve grid configurations per model, searched by grid search with a three-fold chronological `TimeSeriesSplit` and mean absolute error as the selection criterion. Hyperparameters are re-tuned at every forecast origin using only data available at that origin, so that no configuration selected on a later window is transferred to an earlier one. Table 2 documents the complete search space.

Table 2: Hyperparameter search spaces and fixed settings.

Model	Search space / fixed settings	Configurations
Ridge	$\alpha \in \{10^{-3}, \dots, 10^{3}\}$, 8 values log-spaced	8
FFN	hidden layers $\in \{(64, 32), (128, 64)\}$; ℓ_2 penalty $\in \{10^{-4}, 10^{-3}\}$; Adam, learning rate 10^{-3}; early stopping, patience 20 epochs, max. 500	4
XGBoost	max. depth $\in \{3, 4, 5\}$; learning rate $\in \{0.05, 0.1\}$; subsample $\in \{0.7, 0.9\}$; 50 trees during search, 100 at final fit	12
TFT	fixed: hidden size 32, 4 attention heads, dropout 0.1, continuous hidden size 16, Adam, learning rate 10^{-3}, batch size 64, max. 40 epochs, early stopping patience 5, MAE loss	—
Shared: $L = 100$, $H = 100$, $S = 5$ ensemble members, robust scaling fitted on training window, seed 42.		

4.4 Covariate Specifications

To compare how different types of economic information affect forecast accuracy, each non-naive model is evaluated with the nine covariate sets listed in Table 3. These include grouped economic variables, individual interest-rate variables, and subsets of the full predictor set. For Ridge, the FFN and XGBoost, each covariate is supplemented with lags of 1, 5 and 20 calendar days, as defined in Equation (1). The TFT instead uses the selected covariates as sequential inputs and does not require these additional lagged features. The "Code label" column gives the identifier used in the code and figure filenames.

Table 3: Covariate specifications.

Specification	Included variables	Code label
All	€STR, deposit rate, geopolitical risk, inflation, DAX, and ten-year German government bond yield	`alle`
Interest rates	€STR and deposit rate	`nur_zinsen`
Financial markets	DAX and ten-year government bond yield	`nur_maerkte`
Macroeconomic	Geopolitical risk and inflation	`nur_makro`
Rates and macroeconomy	€STR, deposit rate, geopolitical risk, and inflation	`zinsen_makro`
Excluding DAX	All variables except DAX	`ohne_dax`
Excluding inflation	All variables except inflation	`ohne_inflation`
€STR only	€STR	`nur_estr`
Deposit rate only	Deposit rate	`nur_einlagezins`

4.5 Rolling-Origin Evaluation

Forecasting performance is evaluated using an expanding-window rolling-origin design. At each origin o_k, all observations available up to that date form the training sample, and the following $H = 100$ days form the evaluation window:

$$\mathcal{T}_k = \{1, \dots, o_k\}, \qquad \mathcal{E}_k = \{o_k + 1, \dots, o_k + H\}.$$

Models and hyperparameters are re-estimated at every origin. The four origins are 1 January 2009, 1 January 2013, 1 March 2019, and 1 June 2025, representing the financial crisis, the low-interest-rate period, the pre-COVID period, and the post-2022 tightening cycle. The 2009 origin has the shortest training history.

4.6 Evaluation Metrics

Performance is evaluated for both the predicted first differences and the reconstructed deposit-volume levels. Let $y_{k,h}$ and $\widehat{y}_{k,h}$ denote the observed and predicted volume at horizon h after origin o_k. The metrics are

$$\mathrm{MAE}_k = \frac{1}{H}\sum_{h=1}^{H} \left| y_{k,h} - \widehat{y}_{k,h} \right|,$$

$$\mathrm{RMSE}_k = \sqrt{\frac{1}{H}\sum_{h=1}^{H} \left(y_{k,h} - \widehat{y}_{k,h} \right)^2},$$

$$R_k^2 = 1 - \frac{\sum_{h=1}^{H} \left(y_{k,h} - \widehat{y}_{k,h} \right)^2}{\sum_{h=1}^{H} \left(y_{k,h} - \overline{y}_k \right)^2},$$

where $\overline{y}_k$ is the mean observed volume in evaluation window k. Negative R^2 values indicate performance below the test-window mean benchmark.

A scale-free measure. Absolute MAE is not directly comparable across forecast origins because the deposit stock increases over time. We therefore scale each model's MAE by the constant-level benchmark at the same origin:

$$\mathrm{relMAE}_k = \frac{\mathrm{MAE}_k^{\text{model}}}{\mathrm{MAE}_k^{\text{constant-level}}}. \tag{4}$$

Values below one indicate improvement over the benchmark (Hyndman and Koehler, 2006). Metrics are reported as mean and standard deviation across the four origins, with origin-level results in Appendix A. Because four origins provide little power for formal forecast-comparison tests, all comparisons are descriptive.

5 Results

5.1 Forecasting Performance

Table 4 reports the forecasting performance of each model. The covariate specification with the lowest mean level MAE across the four origins, together with the scale-free measure of Equation (4) is specified.

Ridge achieved the lowest average MAE and was the only approach with a positive mean R^2. The TFT ranked second, followed by the one-step drift benchmark. Neither the FFN nor XGBoost improved on that simple benchmark on average, and the constant-level forecast produced the largest errors. Greater model complexity therefore did not consistently translate into higher forecasting accuracy.

Answering RQ1. Forecast accuracy is determined mainly by the relMAE, since absolute errors are affected by the level of the deposit stock. Over the 100-day horizon, Ridge records an error equal to 44% of the constant-level benchmark. The one-step drift benchmark records 60%, while XGBoost and the FFN record 69% and 71%, respectively. Thus, all methods improve on

Table 4: Forecasting performance across the four rolling origins (2009, 2013, 2019, 2025). With MAE and RMSE measured in EUR billions and reported as mean ± standard deviation across origins.

Model	Best covariate set	MAE	RMSE	Mean R^2	relMAE
Estimated models					
Ridge	All covariates	9.36 ± 11.11	10.26 ± 12.08	0.178	0.44 ± 0.31
TFT	Macroeconomic	10.78 ± 12.61	12.71 ± 14.18	−0.361	0.49 ± 0.39
FFN	Macroeconomic	11.85 ± 6.90	15.22 ± 10.35	−0.605	0.71 ± 0.34
XGBoost	Excluding inflation	12.71 ± 13.28	14.90 ± 14.79	−0.784	0.69 ± 0.64
Naive benchmarks					
One-step drift	None	11.78 ± 12.81	14.41 ± 14.25	−0.564	0.60 ± 0.43
Constant level	None	17.38 ± 10.13	21.03 ± 10.52	−2.246	1.00

Note: MAE and RMSE are calculated on reconstructed deposit-volume levels. relMAE is the MAE relative to the constant-level benchmark at the same origin; values below one indicate improvement over that benchmark.

the constant-level benchmark, although Ridge's improvement over the one-step drift forecast is smaller. Figure 4 also shows that the standard-deviation bars overlap almost completely, which is the visual form of the statistical caveat stated in Section 4.6: with four origins, this ranking describes a sample, not a population.

figures/model_comparison.jpeg

Figure 4: Mean MAE of reconstructed deposit-volume levels across the four forecast origins (2009, 2013, 2019 and 2025). Each model is shown using its lowest-MAE covariate specification, and error bars indicate one standard deviation across origins. The horizontal line marks the one-step drift benchmark.

5.2 Accuracy Versus Robustness

Figure 5 compares forecast accuracy and stability for all model-covariate combinations. Ridge has the most specifications lying on the lower-left region, indicating both lower mean MAE and lower dispersion than the one-step drift benchmark. Ridge is the most accurate model across covariate specifications, whereas the FFN specification using macroeconomic covariates has the lowest absolute dispersion (±6.90).

However, after normalising errors by the constant-level benchmark, Ridge achieves the lowest dispersion among the selected models (0.31 in relMAE), followed by the FFN (0.34) and XGBoost (0.64). Moreover, XGBoost achieves an average relMAE of 0.69, compared with 0.71 for the FFN.

figures/performance_vs_stability.jpeg

Figure 5: Forecast accuracy and stability across all model–covariate combinations. The horizontal axis shows mean level MAE across the four forecast origins, while the vertical axis shows its standard deviation. Lower values indicate greater accuracy and stability. The dashed lines mark the mean MAE and standard deviation of the one-step drift benchmark.

This shows why absolute errors are not directly comparable across forecast origins with different deposit levels.

5.3 Effect of Covariate Selection

The preferred covariate specification differs across models, as reported in Table 4. The complete results for all nine specifications are presented in Figure 10 in the appendix.

Ridge is comparatively stable, with mean MAE ranging from 9.36 to 11.81 EUR bn across the nine specifications. This corresponds to an increase of 26.2% from its best to its worst specification. XGBoost shows a similar increase of 26.3%, with mean MAE ranging from 12.71 to 16.05 EUR bn. The TFT is more sensitive, ranging from 10.78 to 14.93 EUR bn, an increase of 38.5%. The FFN is the most sensitive to the selected covariates, with mean MAE ranging from 11.85 to 21.14 EUR bn, an increase of 78.4%.

Ridge's relatively narrow range is consistent with the effect of ℓ_2 regularization, which shrinks the coefficients of weakly informative predictors towards zero. For Ridge, the FFN and XGBoost, changing the covariate specification also changes the size of the flattened input vector. With a 100-day lookback and the current value together with lags of 1, 5 and 20 days, each additional covariate contributes 400 input values. This increase in dimensionality may partly explain the greater sensitivity of the FFN. The TFT processes the covariates sequentially and is therefore not affected by the flattened input dimension in the same way.

Overall, the results show that adding more predictors does not necessarily improve forecast accuracy. The selected specifications describe predictive performance within this sample and should not be interpreted as evidence of causal relationships between the covariates and deposit volumes..

5.4 Variation Across Forecast Origins

Forecast accuracy varies considerably across the four evaluation periods. Over the six methods reported in Table 4, the average mean level MAE is approximately 27.9 Bn. EUR at the 2009 origin, compared with 3.6, 8.3, and 9.4 Bn. EUR at the 2013, 2019, and 2025 origins, respectively. The 2009 period accounts for a substantial part of the dispersion in the aggregate results. This origin combines the global financial crisis with the smallest available training sample. Given the $L = 100$ lookback and $H = 100$ forecast horizon, the effects of the economic regime and limited training history cannot be separated completely.

figures/mae_per_covariate_per_time.jpeg

Figure 6: Level MAE across covariate specifications, separated by model and forecast origin. Each panel represents one model, while the colours identify the four forecast origins. Lower values indicate better forecast performance.

Both figures 6 and 7 show that no model dominates across all forecast origins. The FFN records the lowest error in 2009, Ridge performs best in 2013 and remains competitive in 2019 and 2025, while the TFT records the lowest error at the 2025 origin. Moreover, the best-performing covariate specification changes across origins for every model. The aggregate rankings should then be interpreted as average performance across different economic periods, rather than as evidence that one model or covariate specification is consistently superior.

5.5 Interpreting the Selection Procedure

Each estimated model is reported using the best of nine covariate sets selected on the same four origins used for evaluation, which may overstate its advantage over the untuned naive benchmarks. The origins were chosen deliberately rather than sampled randomly, and standard deviations based on four observations are imprecise. The results therefore do not establish universal Ridge figures/covariate_benchmark_rolling_timeline.png

Figure 7: Rolling-origin evaluation: level MAE (EUR billions, vertical axis) per model plotted against the end of the training window (horizontal axis). Naive benchmarks are drawn as dashed reference lines, estimated models as solid lines with markers.

figures/covariate_benchmark_rolling_summary.png

Figure 8: Distribution of level MAE (EUR billions, vertical axis) across the four rolling origins per model, shown as boxplots ordered by median. Box height reflects the interquartile range, whiskers the full observed range across origins.

superiority, but they provide no evidence that greater complexity consistently improves forecast accuracy.

6 Discussion

The empirical findings from our rolling-origin benchmark provide insight into how distinct algorithmic architectures handle a highly autocorrelated, noisy, and regime-shifting financial series over a 100-day horizon. Contrary to common expectations in the applied machine-learning literature – but consistent with the competition evidence of Makridakis et al. (2020) and Makridakis et al. (2022) and with the linear-baseline results of Zeng et al. (2023) – a simple regularized linear model was not outperformed by any of the more complex architectures. This section evaluates the structural reasons for that outcome, connects the results to the economic events in the sample, and states the limitations that bound our conclusions.

6.1 Ridge Regression: Accuracy Through Shrinkage

Ridge achieved the highest overall accuracy (MAE = 9.36 Bn. €, relMAE = 0.44), outperforming the one-step drift benchmark by 20.5%, and was the only model with a positive mean R^2 (0.178). Under the scale-free measure it was also the most stable of the estimated models (± 0.31), so the phrase "accurate but volatile" that a reading of the absolute standard deviations would suggest does not survive normalisation.

The structural advantage of ℓ_2-regularized linear models in cumulative horizon forecasting lies in their smooth penalty mechanism. Because the level forecast is reconstructed by summation (Equation (3)), a small but *systematic* bias in the predicted daily differences accumulates linearly in the horizon: a bias of 0.05 Bn. € per day becomes an error of 5 Bn. € at $h = 100$. What matters over a long horizon is therefore not the variance of the daily prediction but the absence of drift in its mean. The Ridge penalty shrinks weakly identified coefficients towards zero and thereby suppresses precisely the kind of spurious directional signal that would compound.

This mechanism is decisive at the input dimensionality implied by $L = 100$. With a 100-step window, six covariates and three lags each, the flattened design matrix contains on the order of 2,500 columns, estimated from roughly 1,350 overlapping sequences at the earliest origin – a regime in which the ordinary least-squares solution is not merely imprecise but ill-posed. Neighbouring columns are moreover nearly perfectly collinear by construction, since a covariate at lag j and at lag $j + 1$ differ by one day of a smoothed series, and €STR, the deposit rate and the ten-year yield move together over most of the sample. Shrinkage is the textbook remedy for exactly this configuration, and its effectiveness explains why Ridge was simultaneously the most accurate model, the least sensitive to the covariate set (Figure ??), and best served by the full specification, whereas an unpenalised or weakly penalised learner must expend degrees of freedom distinguishing regressors that carry the same information.

forecast$_linreg_alle_2009-01-01.pngforecast_linreg_alle_2013-01-01.pngforecast_linreg_alle_2019-03-01.p$

The residual sub-panels of Figure ?? make this mechanism visible. At the 2013, 2019 and 2025 origins the residuals remain centred near zero and grow only slowly with the horizon, which is the signature of an unbiased difference forecast. At the 2009 origin the residual panel instead shows a monotone, one-sided drift – the model is not noisy there, it is systematically wrong about the direction of the level, and the summation in Equation (3) converts that directional error into the 25.71 Bn. € error reported in Appendix A.

6.2 Temporal Fusion Transformer: Architectural Capacity Without Input Variation

The TFT ranked second overall (MAE = 10.78 Bn. €, relMAE = 0.49) and produced the single most accurate forecast of any model at any origin (2.43 Bn. € in 2025). Although the TFT is designed specifically for multi-horizon forecasting through variable selection and self-attention (Lim et al., 2021), its potential was structurally constrained by the evaluation setup.

To prevent look-ahead bias, future values of the six macro-financial covariates were held constant at their last observed value over the entire horizon; only the calendar features were supplied as genuine known future inputs. The decoder therefore receives a constant sequence for every economically meaningful variable. The attention mechanism, whose purpose is to weight informative future inputs against one another, has nothing to weight. This is not a defect of the architecture but a property of the forecasting problem: for a bank forecasting at time t, the future path of inflation and market rates is genuinely unknown, and any protocol that supplied it would report accuracy that could not be realised in practice. It does, however, mean that our comparison evaluates the TFT under conditions that neutralise its main design advantage, and the conclusion should be read as "the TFT offers no advantage *when future covariates are unavailable*" rather than as a general statement about the architecture. Notably, the 100-day encoder gives the TFT a genuinely long context over which attention can operate on the historical side, so the constraint lies unambiguously on the decoder input rather than on the amount of history available.

forecast$_tft_nur_makro_2009-01-01.pngforecast_tft_nur_makro_2013-01-01.pngforecast_tft_nur_makro_2019-$

Figure **??** shows the consequence: with frozen exogenous inputs, the predicted path degenerates towards a smooth extrapolation of the encoder trend, and the ensemble bands widen steadily with the horizon as the model signals its own growing uncertainty.

6.3 Feedforward Networks and Gradient-Boosted Trees

Neither the FFN (11.85 Bn. €) nor XGBoost (12.71 Bn. €) consistently outperformed the one-step drift benchmark (11.78 Bn. €), despite both having been tuned under the same protocol and budget as Ridge (Section 4.3). The two models fail in structurally different ways.

- **FFN: consistency without precision.** The FFN exhibited the lowest absolute standard deviation across origins (±6.90) and produced the best forecast of any model at the difficult 2009 origin (20.18 Bn. €). Its weakness is concentrated in the easy windows: at the 2013 origin, where Ridge achieved 1.00 Bn. € and the TFT 1.05, the FFN required 3.36. In other words, it behaves like a heavily damped forecaster – it never fails catastrophically, but it also never converges tightly on the target when the target is easy. Part of this behaviour is attributable to the input dimensionality: a dense first layer over a flattened 100-step window must estimate a weight for every time step of every covariate, and with a ℓ_2 penalty of at most 10^{-3} that is far weaker than the value Ridge selects, the network has little protection against fitting noise in individual lag positions. For a liquidity manager whose concern is tail behaviour rather than average behaviour, this profile is nonetheless not without value, a point we return to below.

- **XGBoost: the extrapolation trap.** Regression trees partition the feature space into orthogonal hyper-rectangles and predict a constant within each leaf. They therefore cannot extrapolate a trend beyond the range of differences observed in training. This limitation is most severe exactly where deposit dynamics left their historical range: at the 2025 origin, following the largest reallocation of household deposits since the introduction of the euro, XGBoost recorded 15.77 Bn. € against 6.05 for Ridge and 2.43 for the TFT. XGBoost is also the least stable model under the scale-free measure (±0.64), which is the same fact seen from another angle.

Figure 9 contrasts the two failure modes directly: the FFN produces a smooth path with a persistent level offset, whereas the XGBoost path flattens after a small number of steps and the residual panel accumulates an almost monotone error across the remainder of the horizon.

figures/forecast_ffn_nur_makro_2025-06-01.png

(a) FFN, macroeconomic covariates (best specification)

figures/forecast_xgboost_nur_einlagezins_2025-06-01.

(b) XGBoost, deposit rate only

Figure 9: Forecasts of the two nonlinear models at the 1 June 2025 origin, the window most affected by the post-2022 deposit reallocation. Panels are constructed as in Figure **??**. Note that the XGBoost panel shows the *deposit-rate-only* specification rather than the best-performing "excluding inflation" specification of Table 4; it is displayed here because it isolates the extrapolation behaviour discussed in the text most clearly.

6.4 Economic Interpretation: Why 2009 and 2022 Are Hard

The two hardest parts of our sample correspond to the two episodes in which household deposit behaviour was driven by forces that no covariate in our information set measures.

The January 2009 origin sits immediately after the collapse of Lehman Brothers, during a period of acute precautionary saving, deposit guarantees and flight to safety. Deposit volumes moved for reasons of confidence rather than of relative return, and the resulting path is not a function of interest rates, equity prices or measured inflation over the preceding hundred days. Every model, including both naive benchmarks, records its largest error at this origin – the crisis is not a model-selection problem but an identification problem, compounded by the fact that this origin also offers the least training data (Section 4.1).

The second episode shapes the 2025 origin. The 2020–2021 pandemic produced a large accumulation of household overnight balances through forced saving, and the ECB tightening cycle from mid-2022 then triggered the reverse: a substantial reallocation from overnight into term deposits as the spread between them widened for the first time in over a decade. This is a behavioural response to a *spread* and to a change in the speed of deposit-rate pass-through, neither of which appears as a variable in Table 3 – we include the deposit rate and €STR separately, but not their difference and not the pass-through coefficient between them. That the TFT nonetheless achieved 2.43 Bn. € at this origin while XGBoost required 15.77 reflects the extrapolation argument above rather than superior economic understanding. Constructing an explicit deposit-beta or spread covariate is, in our view, the single most promising extension of this work, and we return to it in Section 7.

6.5 Understanding the Out-of-Sample R^2 Metric

As observed in Table 4, most models yielded negative R^2 values. In multi-step time-series forecasting, R^2 compares predictions against the mean of the test window, $\overline{y}_k$. Because $\overline{y}_k$ is computed from realizations that are unavailable at the forecast origin, it is an unrealistically strong reference: a forecaster who knew the average deposit level over the coming quarter would already possess most of the information being forecast. A negative R^2 therefore indicates that the multi-step trajectory diverged from this post-hoc mean, not that the model has no predictive utility – as the relMAE column confirms, every model beats the genuinely available constant-level reference. Level MAE and relMAE are therefore our primary ranking metrics, and R^2 is reported for comparability with prior work only.

6.6 Limitations

Several features of this study bound the conclusions that can be drawn from it, and we set them out explicitly.

The target is an interpolated series. This is the most consequential limitation. The deposit volume is published monthly and converted to a daily grid by PCHIP interpolation (Section 3.2). Two consequences follow. First, an interpolated value within a month is constructed using the value at the end of that month, which was not known in real time; the constructed "truth" within the evaluation window is therefore partly forward-looking. Second, the resulting daily differences are artificially smooth and strongly autocorrelated, which flatters extrapolative methods – including our one-step drift benchmark – and means that the 100 error terms entering each MAE_k represent roughly three independent observations rather than one hundred. All reported metrics should be read as describing forecast accuracy against a smoothed construct, not against realised daily balances, and a replication at native monthly frequency is the first robustness check we would recommend.

Few origins, and covariate sets selected in-sample. Four forecast origins are too few for inference, and the best covariate set per model is chosen on the same windows on which it is reported. Both issues are analysed in Section 5.5; we report descriptively throughout and avoid claims of statistical superiority.

The tuning budget is equal but not equally generous. Ridge, XGBoost and the FFN were tuned under an identical protocol and an identical budget of at most twelve configurations. The TFT architecture was fixed a priori, because a comparable search was not computationally feasible. A more extensively tuned TFT might close the remaining gap to Ridge. This asymmetry works against our conclusion, but it means that our result is properly stated as "complexity did not pay off *at this level of tuning effort*".

The lookback window is fixed and not varied. We set $L = H = 100$ for all models on the grounds given in Section 4.1 and did not vary it. Two consequences follow. First, a window of 100 trading days does not span an annual cycle of approximately 250 trading days, so no model can learn calendar seasonality or interest-rate cycles from its encoder input; monthly and quarterly effects are in any case largely removed by the interpolation of the target. Second, the requirement of $L + H = 200$ observations per training sequence materially reduces the effective sample, and does so most severely at the earliest origin, which is also the hardest. Both effects apply identically to all models and therefore do not bias the comparison between them, but they bound the level of accuracy that is attainable in this design, and it remains possible that the flexible architectures would separate from Ridge given a substantially longer context. A sensitivity analysis over $L \in \{100, 250, 500\}$ would test this directly and is, alongside the monthly-frequency replication, the most important robustness check left open by this study.

Training and inference see different covariate structures. Models are trained on windows in which covariates vary, but forecast with covariates frozen at the origin. This mismatch penalises precisely those models that rely most heavily on covariate dynamics and offers a plausible alternative explanation for why the FFN and TFT achieve their best results with the smallest covariate set. Our RQ3 findings should therefore be read as statements about behaviour under this realistic-information protocol, not as evidence about the intrinsic economic informativeness of the variables.

Aggregate data cannot answer institution-level questions. Our motivation is institution-level ALM, but our data is a sector aggregate published by the Bundesbank. An aggregate series nets out migration between institutions, contains no information on customer segmentation, and reveals neither the deposit beta of an individual bank nor its core/non-core split – which are precisely the quantities that replicating-portfolio and IRRBB frameworks require (Basel Committee on Banking Supervision, 2016; European Banking Authority, 2022; Frauendorfer and Schürle, 2003). Our results are therefore best understood as evidence about the predictability of *sector-level* household deposit dynamics, which is useful as a macro-prudential and early-warning input, and as methodological guidance on model selection that plausibly transfers to institution-level data. They are not a validated forecasting system for an individual bank's deposit book.

Uncertainty bands are not calibrated. The 90% and 98% bands in Figures **??** to 9 are empirical quantiles across ensemble members and therefore quantify estimation uncertainty, not total forecast uncertainty; they are consequently too narrow, and no coverage evaluation is reported. Since liquidity buffer sizing depends on interval accuracy rather than on point accuracy, this is a substantive gap.

Reconstructed €STR history. For observations before March 2017 the €STR series is reconstructed rather than observed, following Winkler (2025). A substantial portion of the sample therefore contains a synthetic version of one of our central covariates.

6.7 Managerial Implications

For bank asset–liability management and treasury operations, predictability, computational cost, and interpretability are as consequential as raw accuracy. Three implications follow from our results.

First, a regularized linear model is a defensible default for 100-day aggregate deposit forecasting. It was not outperformed by any more complex architecture, its coefficients can be inspected and challenged by a model validation function, it trains in seconds rather than in the minutes to hours required by the TFT, and it is straightforward to document under model-risk governance requirements.

Second, and more importantly, the marginal value of *any* of these models over a naive rule is modest compared with the value of having a model at all. The gap between the best model and the one-step drift rule (relMAE 0.44 against 0.60) is far smaller than the gap between either and a constant-level assumption (1.00). Institutions currently without a volume model gain most from adopting a simple one; institutions with a simple one should not expect large gains from replacing it with a deep architecture.

Third, model choice should follow the intended use. For central-case planning, the lowest average error is the relevant criterion and favours Ridge. For stress and buffer sizing, what matters is behaviour in the adverse regime, and there the picture differs: at the 2009 crisis origin the FFN was the most accurate model of all, and it exhibited the smallest absolute dispersion across regimes. A treasury function may therefore rationally run a simple model for planning alongside a more heavily damped one for tail-scenario assessment, rather than searching for a single winner.

7 Conclusion and Outlook

Accurate forecasting of non-maturity deposit volumes is fundamental to bank liquidity management, regulatory balance-sheet planning, and interest-rate risk assessment. Modelling aggregate NMDs is nonetheless difficult because the series is non-stationary, strongly autocorrelated, and subject to macroeconomic regime shifts, and because the existing literature treats the volume process largely as a valuation input rather than as a forecasting target evaluated out of sample.

This study evaluated six forecasting approaches across four forecast origins spanning distinct interest-rate regimes (2009–2025) using an expanding-window rolling-origin design, a common information set with a 100-day lookback window and lagged covariates, and a common tuning budget. Our conclusions address the three research questions as follows.

1. **Forecast accuracy (RQ1).** Over a 100-day horizon, the best model reduced the mean absolute error to 44% of the constant-level benchmark (MAE = 9.36 Bn. €), averaged across four very different economic environments. Accuracy varied far more across periods than across models: the 2009 crisis origin was roughly three times harder than any other and accounts for nearly all reported cross-origin dispersion.

2. **Model complexity (RQ2).** Greater complexity did not yield higher accuracy. Ridge achieved the best overall performance (relMAE = 0.44, mean $R^2 = 0.178$) ahead of the TFT (0.49), while XGBoost (0.69) and the FFN (0.71) failed to improve on a one-step drift rule (0.60) despite receiving identical inputs, an identical context length, and an identical tuning budget. Because only four origins are available, we report this as an absence of measurable advantage rather than as demonstrated superiority of the linear model.

3. **Covariate selection (RQ3).** The value of a covariate set depends more on the functional form that consumes it than on its economic content. Ridge varied by only 26% across the nine specifications and was best with the full set, consistent with shrinkage handling the highly collinear design matrix that a 100-step window produces; the FFN varied by a factor of nearly two and was best with the smallest set. Since the optimal set also differs by origin, we treat these as sample-specific optima rather than transferable recommendations.

Outlook and Future Work. Four extensions follow directly from the limitations set out in Section 6.6. First, and most urgently, the benchmark should be replicated at the native monthly frequency of the target, so that accuracy is measured against realised rather than interpolated values. Second, the lookback window should be varied systematically over $L \in \{100, 250, 500\}$: a context spanning at least one full year would allow calendar and cyclical structure to be learned, and would test whether the parity between Ridge and the complex architectures reported here is a property of the task or of the context length we imposed. Third, the frozen-covariate protocol could be replaced by scenario-based path projections for the dynamic covariates, which would restore the TFT's known-future-input mechanism and permit a fair assessment of its design advantage. Fourth, the economic analysis in Section 6.4 suggests that the informative variable for the post-2022 period is not the deposit rate or the market rate individually but the spread between overnight and term deposits together with the speed of pass-through; constructing an explicit deposit-beta covariate is, in our assessment, more promising than any further change of architecture. Finally, hybrid designs that combine a regularized linear trend component with a nonlinear residual model may capture sudden systemic liquidity shocks while preserving the stability that made the linear baseline competitive here.

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A Appendix: Additional Figures and Tables

Table 5: Complete per-origin results for each model in its best covariate specification. Level MAE in EUR billions; relMAE in parentheses is the ratio to the constant-level benchmark at the same origin. With only four origins, we report the full matrix rather than aggregates alone.

Model	Covariate set	2009-01-01	2013-01-01	2019-03-01	2025-06-01
Ridge	All covariates	25.71 (0.79)	1.00 (0.08)	4.67 (0.33)	6.05 (0.57)
TFT	Macroeconomic	28.44 (0.88)	1.05 (0.09)	11.19 (0.78)	2.43 (0.23)
FFN	Macroeconomic	20.18 (0.62)	3.36 (0.28)	12.77 (0.89)	11.08 (1.04)
XGBoost	Excluding inflation	30.20 (0.93)	1.26 (0.10)	3.63 (0.25)	15.77 (1.49)
One-step drift	None	30.23 (0.93)	2.80 (0.23)	3.43 (0.24)	10.68 (1.01)
Constant level	None	32.40 (1.00)	12.20 (1.00)	14.30 (1.00)	10.62 (1.00)
Mean across models	—	27.86	3.61	8.33	9.44

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figures/covariate_results_full.png

Figure 10: Complete forecasting results across the nine covariate specifications. MAE and RMSE are measured in EUR billions and reported as mean ± standard deviation across the four rolling forecast origins. Specifications are ordered by mean MAE within each model, and the best specification for each estimated model is shown in bold.

figures/overview_normalized.png

Figure 11: All variables after standardisation to zero mean and unit variance over the sample period January 2003 – September 2025, shown on a common vertical scale so that co-movement between the deposit volume and the covariates is directly visible.

B Appendix: Code

The complete implementation, including the rolling-origin driver, the tuning protocol of Table 2, and the scripts producing every figure in this paper, is available in the accompanying repository. The excerpt below shows the interface of the shared model-fitting routine.

Listing 1: Shared model-fitting interface.

```
def fit_sequence_model(model_name, X_train, y_train, seed=SEED,
                       bootstrap=False, do_tune=False,
                       tuned_estimator=None):
    """Fit one ensemble member of a sequence model.

    model_name      : "linreg" | "ffn" | "xgboost"
    bootstrap       : resample training sequences (ensemble members S > 1)
    do_tune         : run the shared grid search (<= TUNING_BUDGET configs)
    tuned_estimator : reuse a cached configuration for the same origin
    """
    ...
```